

Maps (Filled) Guided Lab (Part 2)

Robert Thomas Kohn

10/28/25

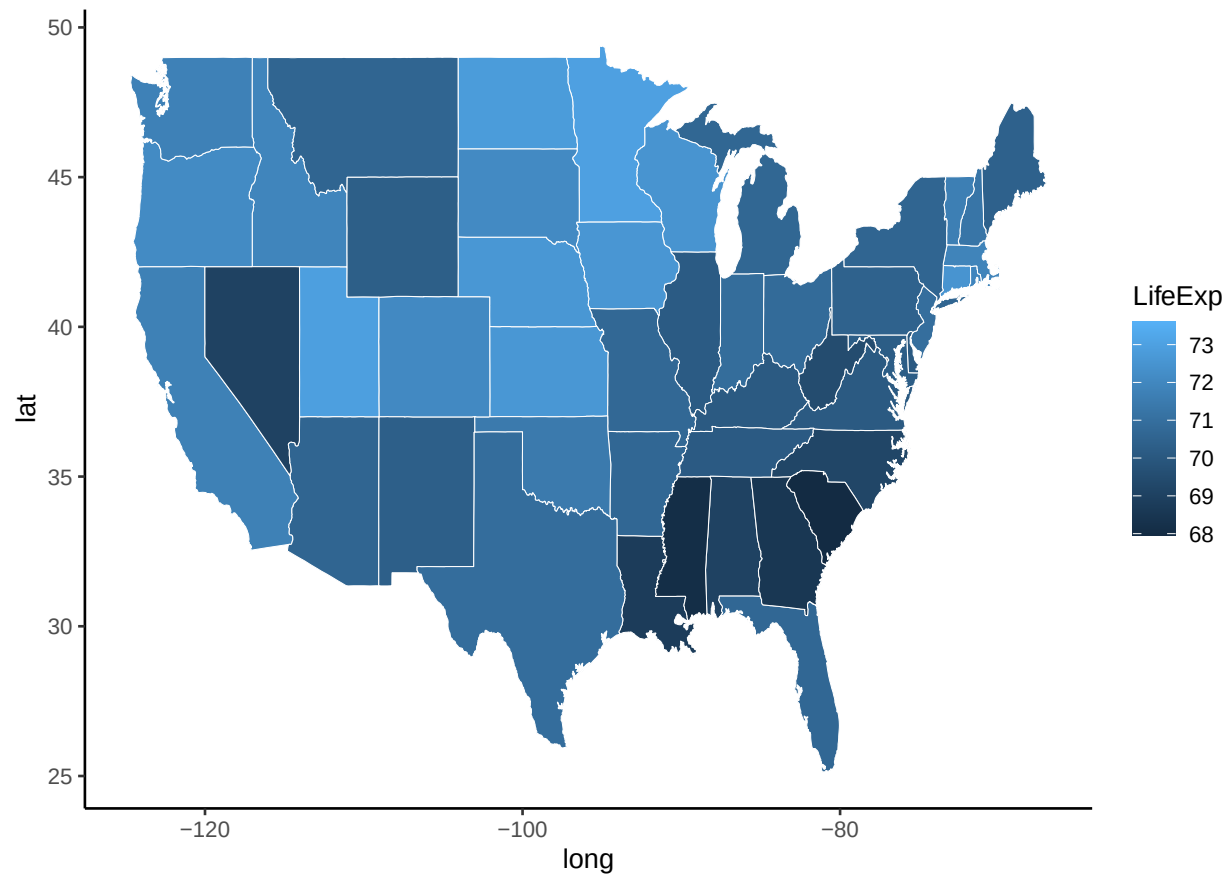
Filling Maps with Color

State abbreviations instead of state names: `mutate( FullState = abbr2state( AbbState ) )` State has upper-case letters and we need lowercase: `mutate( state = tolower( State ) )` Combine Both in one command: `mutate( state = tolower( abbr2state( AbbState ) ) )`

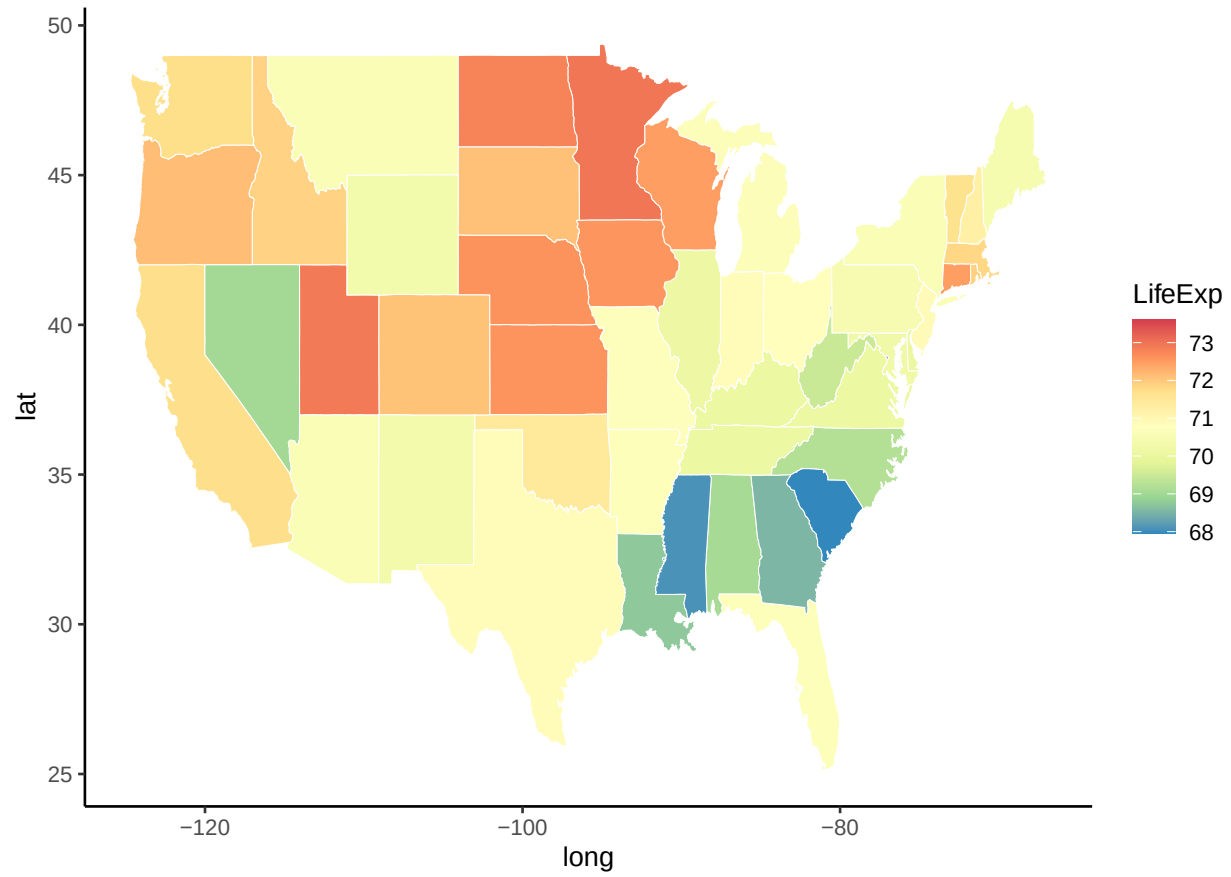
`gf_polygon( Lat ~ Long , data = Dataset , group = ~ Group , fill = ~ Fill , color = " LineColor " , size = LineSize )` Conic for Entire US Map: `coord_map("conic", lat0=30)` Equal Coords for Single State, Country, or World Map: `coord_equal( )` or `coord_fixed( )`

Graph 1: State map filled by numeric variable

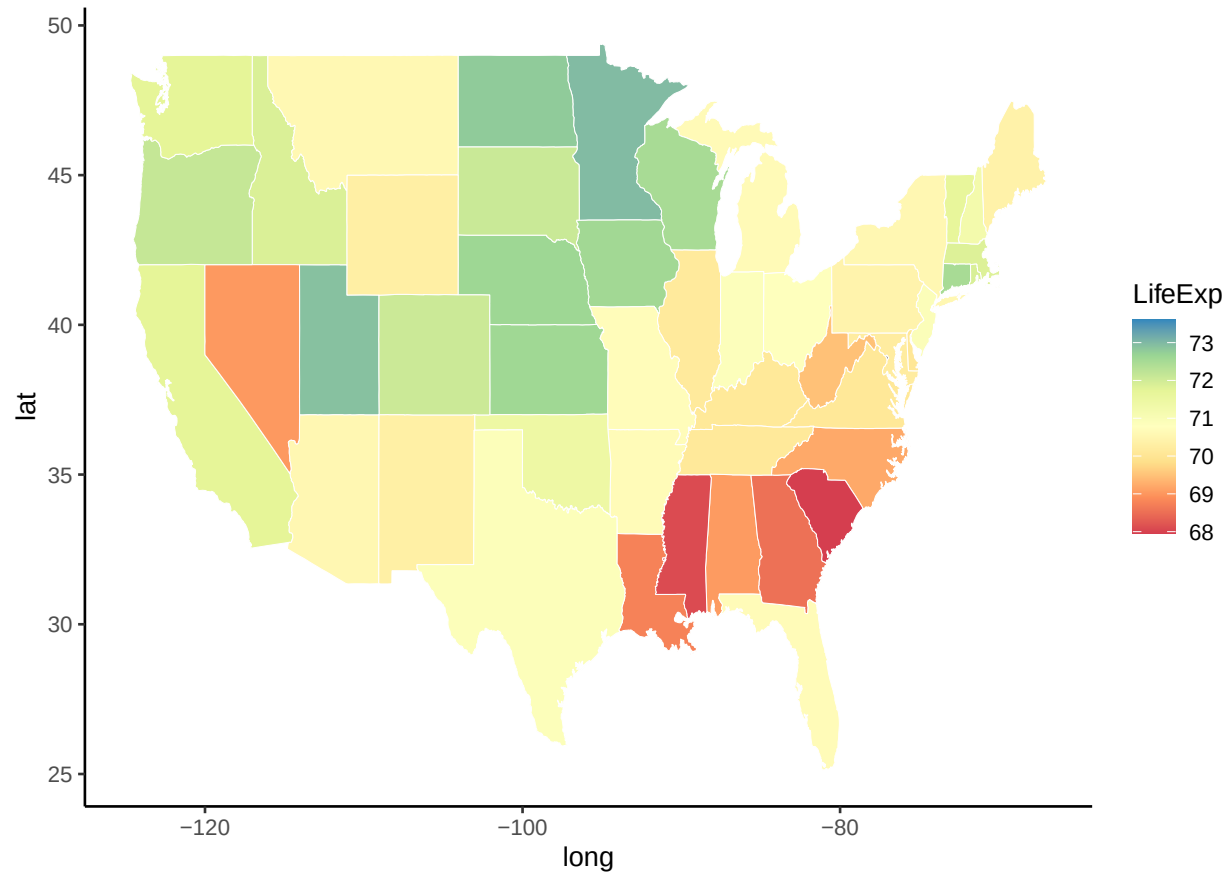
```
NewState77 = rownames_to_column( as.data.frame( state.x77 ) ) %>% # make row names a usable variable
rename(LifeExp = `Life Exp`, # Change messy variable names to clean ones
HSGrad = `HS Grad` ) %>% # Use ` (upper left on keyboard) instead of ' (right middle)
mutate(region = tolower( rowname ) ) # Make region variable with lowercase states
states <- map_data("state")
FullState77 <- full_join(NewState77, states, by = "region")
gf_polygon(lat ~ long, data = FullState77, group = ~group, fill = ~LifeExp, color = "white", size = 0.2)
```



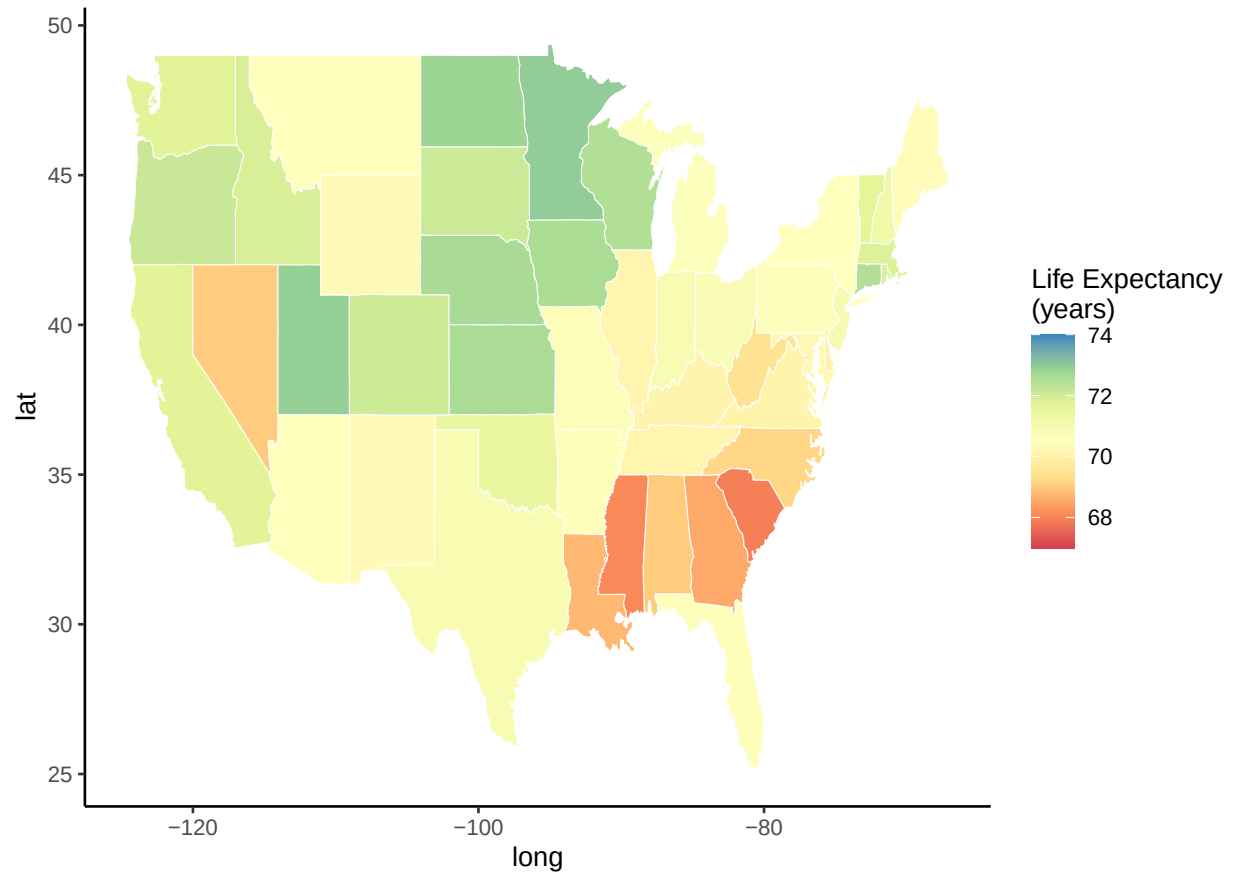
```
gf_polygon(lat ~ long, data = FullState77, group = ~group, fill = ~LifeExp, color = "white", size = 0.2)
```



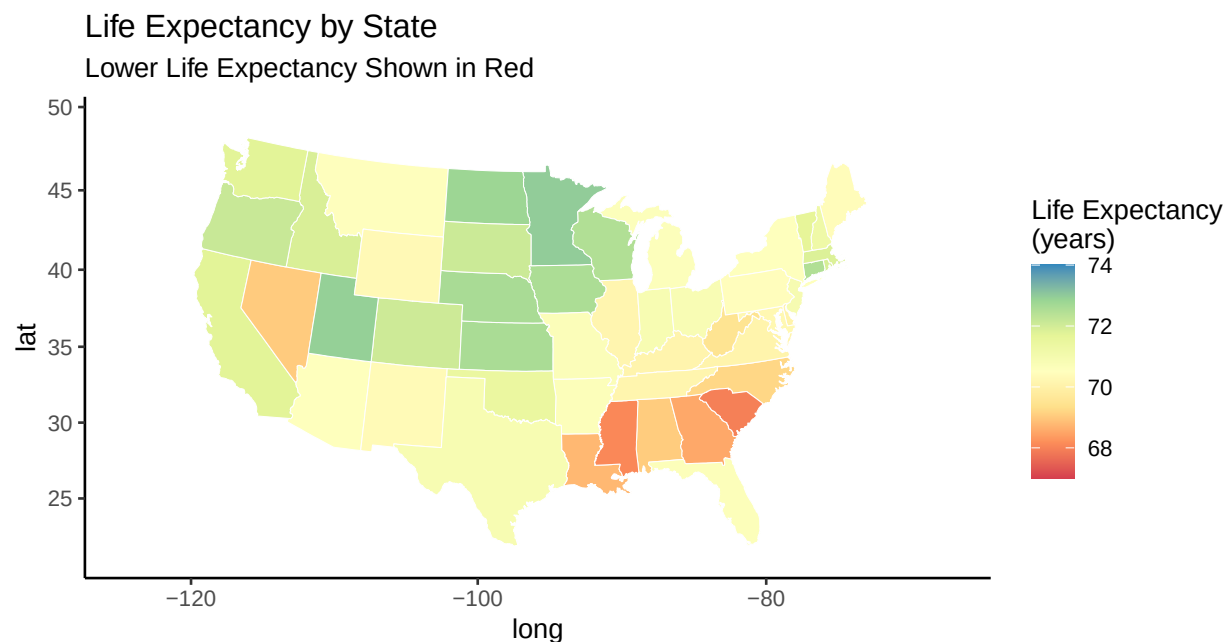
```
gf_polygon(lat ~ long, data = FullState77, group = ~group, fill = ~LifeExp, color = "white", size = 0.2)
```



```
gf_polygon(lat ~ long, data = FullState77, group = ~group, fill = ~LifeExp, color = "white", size = 0.2)
```



```
gf_polygon(lat ~ long, data = FullState77, group = ~group, fill = ~LifeExp, color = "white", size = 0.2)
```



Graph 2: State map filled by categorical variable

```
NewState77 = NewState77 %>%
mutate( binned_life = cut( LifeExp , breaks = 5 , # Create a categorical variable from a numeric one
labels = c( "Really Low" , "Low" , "Medium" , "High" , "Really High" ) ) )
names(NewState77)
```

```
## [1] "rowname"      "Population"   "Income"       "Illiteracy"   "LifeExp"
## [6] "Murder"       "HSGrad"      "Frost"        "Area"         "region"
## [11] "binned_life"
```

```
head(NewState77)
```

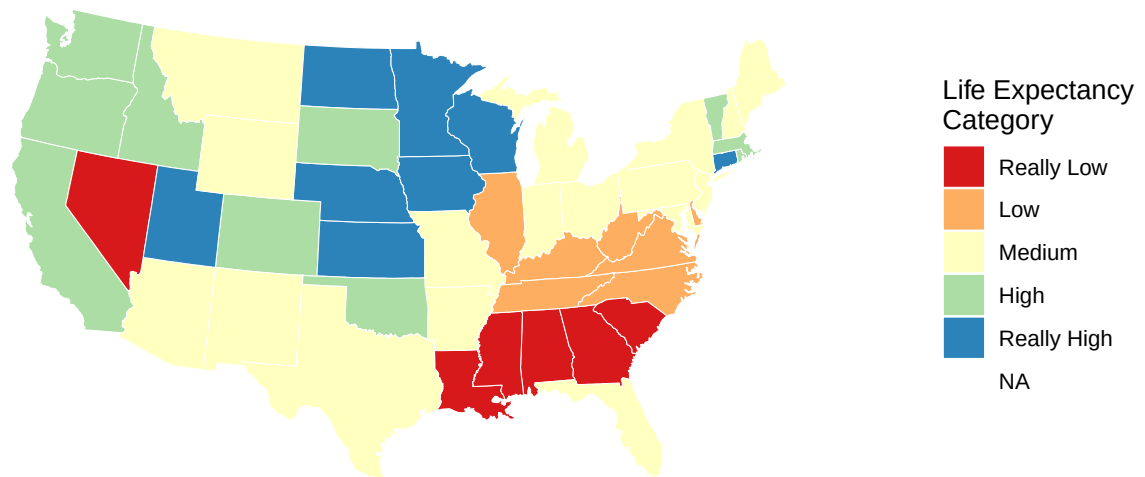
```
##      rowname Population Income Illiteracy LifeExp Murder HSGrad Frost Area
## 1  Alabama      3615   3624        2.1   69.05   15.1   41.3   20  50708
## 2  Alaska        365   6315        1.5   69.31   11.3   66.7   152 566432
## 3  Arizona      2212   4530        1.8   70.55    7.8   58.1    15 113417
## 4  Arkansas      2110   3378        1.9   70.66   10.1   39.9    65  51945
## 5 California    21198   5114        1.1   71.71   10.3   62.6    20 156361
## 6  Colorado      2541   4884        0.7   72.06    6.8   63.9   166 103766
##      region binned_life
## 1  alabama  Really Low
## 2  alaska   Low
## 3  arizona   Medium
## 4  arkansas  Medium
```

```
## 5 california      High
## 6   colorado      High

FullState77 <- full_join(NewState77, states, by = "region")
gf_polygon(lat ~ long, data = FullState77, group = ~group, fill = ~binned_life, color = "white", size = 0
```

Life Expectancy by State (Categorical)

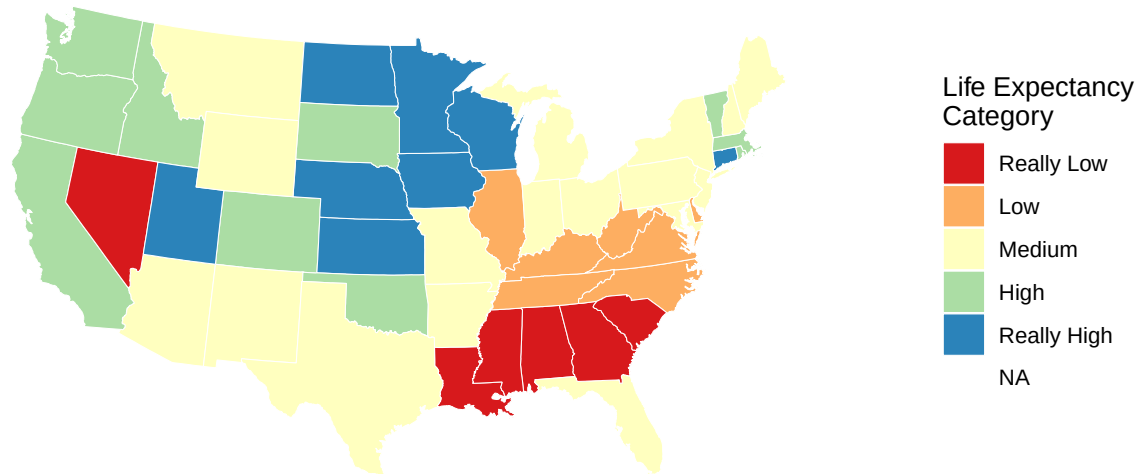
Binned into 5 categories



```
gf_polygon(lat ~ long, data = FullState77, group = ~group, fill = ~binned_life, color = "white", size = 0
```

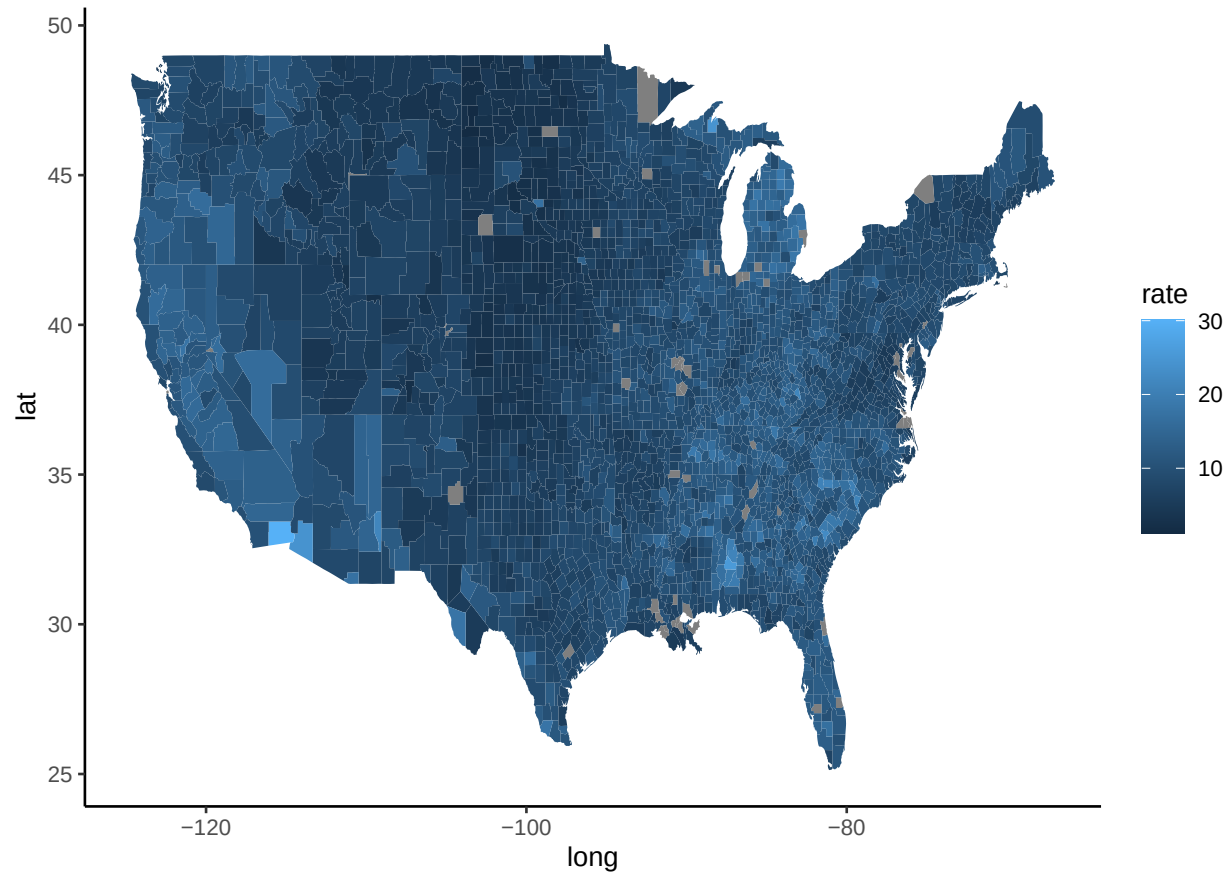
Life Expectancy by State (Categorical)

Binned into 5 categories

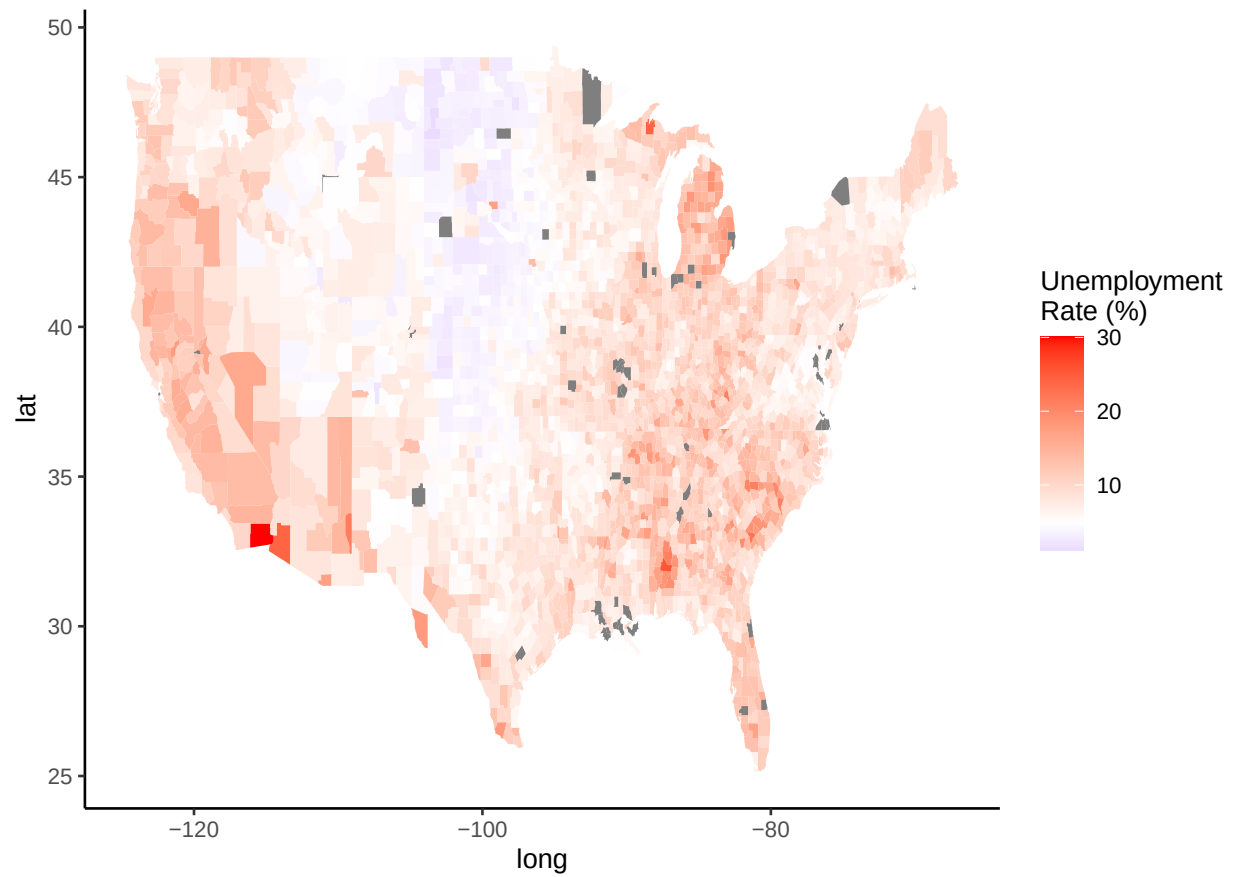


Graph 3: County map filled diverging sequence

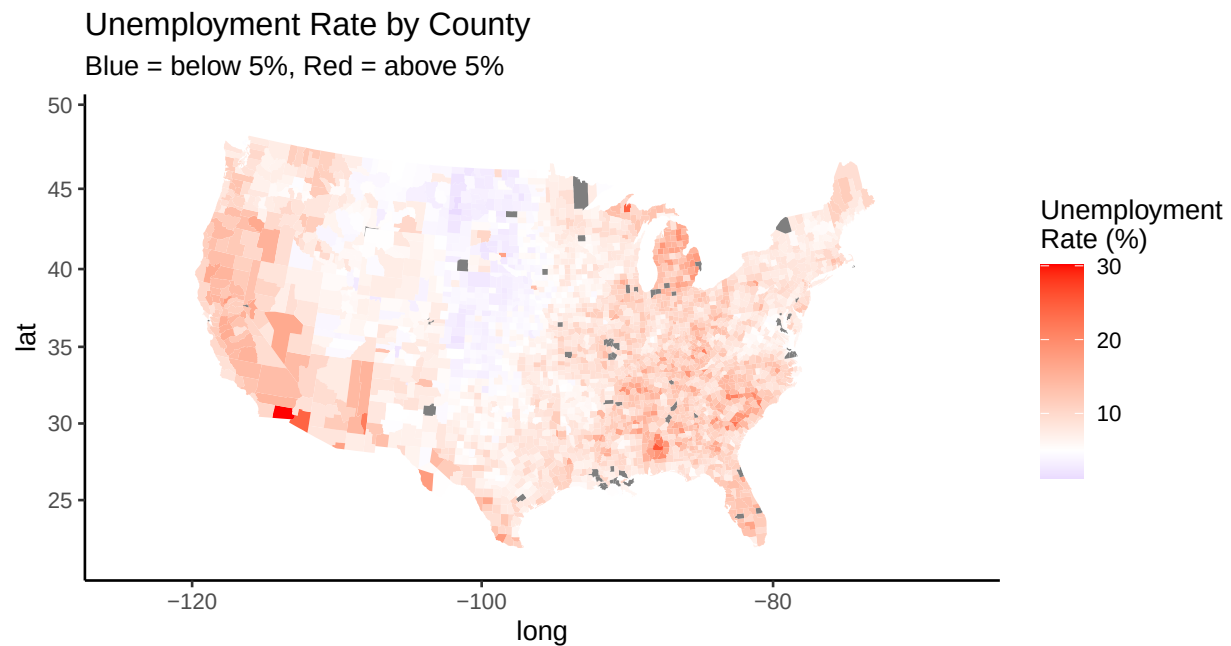
```
county <- map_data("county")
emp2 = viridis :: unemp %>% # Load unemp data from the viridis package
mutate( region = tolower( abbr2state( state ) ) ) # Make abbreviations state names instead
emp.county = full_join( emp2 , county , join_by( region , county == subregion ) )
gf_polygon(lat ~ long,data = emp.county,group = ~group,fill = ~rate, color = NA)
```



```
gf_polygon(lat ~ long, data = emp.county, group = ~group, fill = ~rate, color = NA) %>% gf_refine(scale_fi
```



```
gf_polygon(lat ~ long, data = emp.county, group = ~group, fill = ~rate, color = NA) %>% gf_refine(scale_fi
```

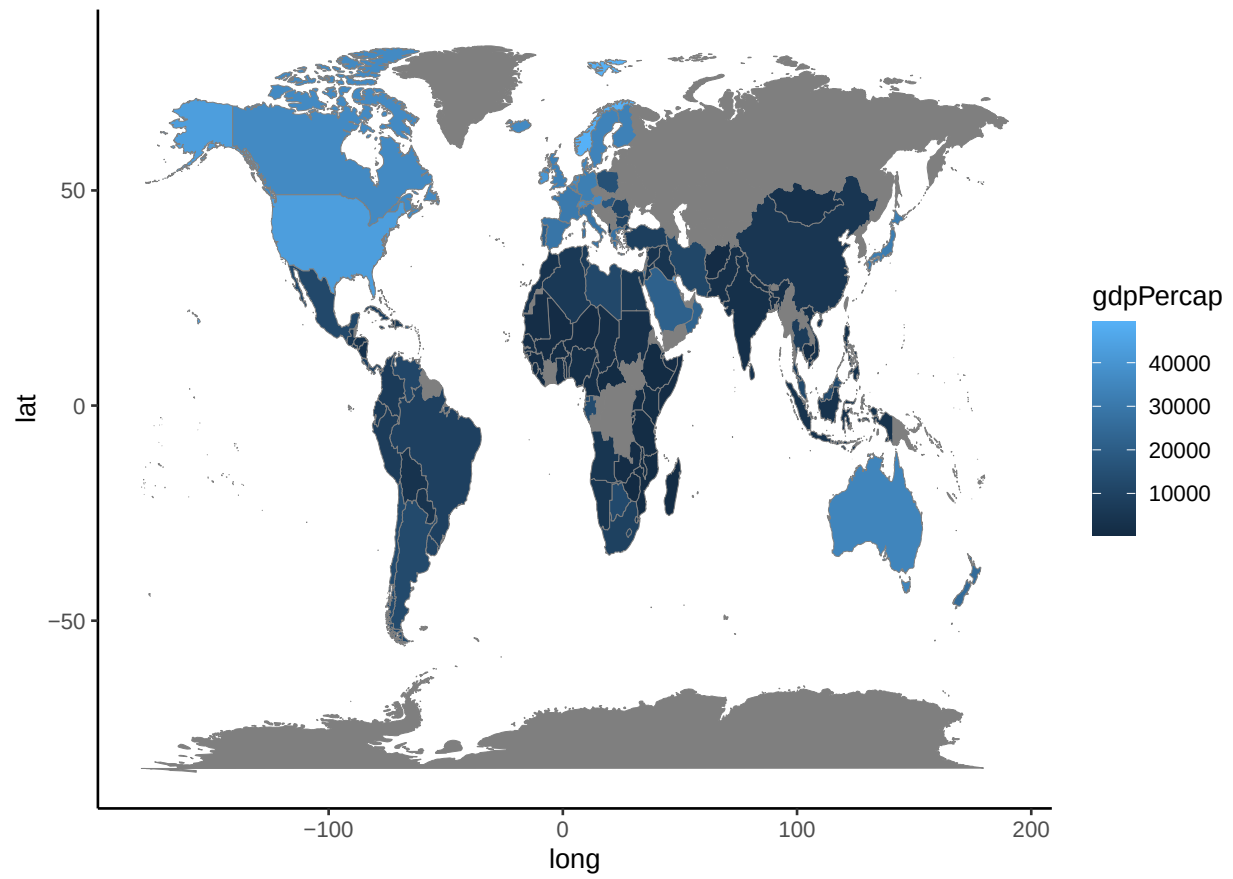


Graph 4: World map filled

```
world <- map_data("world")
gapminder2007 <- gapminder %>%filter(year == 2007)
gapminder.newnames = full_join(gapminder2007 , Countries, join_by( country == gapminder ))
world.gapminder = left_join(world, gapminder.newnames , join_by( region == maps))
```

```
## Warning in left_join(world, gapminder.newnames, join_by(region == maps)): Detected an unexpected many-to-many relationship.
## i Row 97181 of `x` matches multiple rows in `y`.
## i Row 156 of `y` matches multiple rows in `x`.
## i If a many-to-many relationship is expected, set `relationship = "many-to-many"` to silence this warning.
```

```
gf_polygon(lat ~ long,data = world.gapminder,group = ~group,fill = ~gdpPercap, color = "gray50", size = ~pop)
```



```
gf_polygon(lat ~ long, data = world.gapminder, group = ~group, fill = ~gdpPerCap, color = "gray50", size =
```

GDP per Capita by Country (2007)

Log scale transformation

